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Electro-Magnetic Field (force caused by potential):

$$\mathcal{B} = \nabla \times U_m = \frac{1}{c} \left(\frac{r}{|r|} \times \mathcal{E} \right) \quad \text{Near Field} \approx \dots \quad \text{Far Field} \equiv \dots \quad (1)$$

$$\mathcal{E} \stackrel{\text{conservativ}}{=} -\nabla U_e \stackrel{\text{dynamic}}{=} -\nabla U_e - \frac{\partial}{\partial t} U_m \stackrel{\text{fast moving charge}}{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{\varepsilon_0} \frac{1}{r} \frac{1}{\left(1 - \frac{r}{|r|} \cdot \frac{v}{c}\right)^3} q \left[\underbrace{\left(1 - \frac{v^2}{c^2}\right) \frac{\frac{r}{|r|} - \frac{v}{c}}{r}}_{\substack{\text{uniform movement} \\ (2)}} + \underbrace{\frac{\frac{r}{|r|} \times \left[\left(\frac{r}{|r|} \cdot \frac{v}{c}\right)\right]}{\dots}}_{\text{accelerate}}$$